

Change the tool before it scraps the part.

AI-powered predictive maintenance for CNC cutting tools — a physics-grounded wear model and a calibrated risk model, cross-checked against each other, explained per prediction, and validated the way an engineering paper is validated rather than a pitch deck.

Trained on PHM 2010 Data Challenge — real, physically measured tool wear (CC BY 4.0)

Cross-checked with Taylor's tool-life equation, $V \cdot T^n = C$ (1907)

Explained with SHAP, per prediction

KEY INNOVATION

Metrik pairs a machine-learning wear estimate with Taylor's century-old tool-life equation, so every prediction is checked against physics as well as data — then explains itself with SHAP, and closes the loop with an operator-verdict scoreboard, instead of handing over one opaque number.



16.0µm

wear MAE
(leave-one-
tool-out)

0.983

risk model
ROC-AUC

2

independently
validated
models

0

machine
actions —
decision

TOP SHAP DRIVERS



support
only



Process Temperature

Tool wear is invisible until it isn't.

THE PROBLEM

Wear happens where no one can see it.

Flank wear develops at the tool tip, inside the cut, under coolant — an operator cannot see it, and by the time a part shows a defect, the tool has already produced scrap. So most shops fall back on fixed-schedule tool changes: swap every N parts, or every shift, whether the tool needs it or not.

WHY IT MATTERS

A schedule is always wrong in one direction.

Change too early and a shop discards carbide inserts with real cutting life left, inflating tooling spend across hundreds of changes a month. Change too late and a degrading tool drifts out of tolerance, scraps the workpiece, or — at spindle speed — fractures, which is a genuine hazard to the operator standing at the machine, not just a cost.

THE DATA ALREADY EXISTS

It's discarded, not interpreted.

Spindle speed, torque, temperature and cumulative runtime already stream off CNC machines during every cut. The gap isn't sensing — it's turning that stream into a number an operator can act on, and trust.

Existing Approaches → The Gap

Fixed schedule

WHAT IT DOES
Replaces tools on a calendar or part-count interval, no matter their actual condition.

THE GAP
Wastes usable tool life on early changes, and still misses a tool that wears out ahead of schedule.

Vendor monitoring hardware

WHAT IT DOES
Dedicated sensor packages raise an alert when a reading crosses a vendor-set threshold.

THE GAP
Expensive and proprietary; the alert is opaque, so operators can't verify it and stop trusting it.

Two questions, kept deliberately separate.

Metrik treats "will this cut scrap the part?" and "how worn is this insert right now?" as two different questions — because only one of them can be honestly answered from a synthetic benchmark, and conflating them is how monitoring tools end up quoting a number they can't actually support. Keeping them separate, and being explicit about which is which, is itself part of the innovation.

01

Physics-Grounded, Not Black-Box

Every wear estimate is checked against Taylor's tool-life equation ($V \cdot T^n = C$, 1907) — the standard the machining industry already trusts — instead of asking an operator to accept a machine-learning number on faith alone.

02

Validation Methodology as a Feature

The wear model is validated leave-one-tool-out — tested on a cutter it never trained on. Measured on the same data, a naive random split reports a **10.7× more optimistic** error, and that gap is reported, not hidden.

03

Explainable Per Prediction

SHAP decomposes every single prediction into a ranked list of which parameter is driving it — torque, runtime, temperature — so an operator knows what to correct, not just that something is wrong.

04

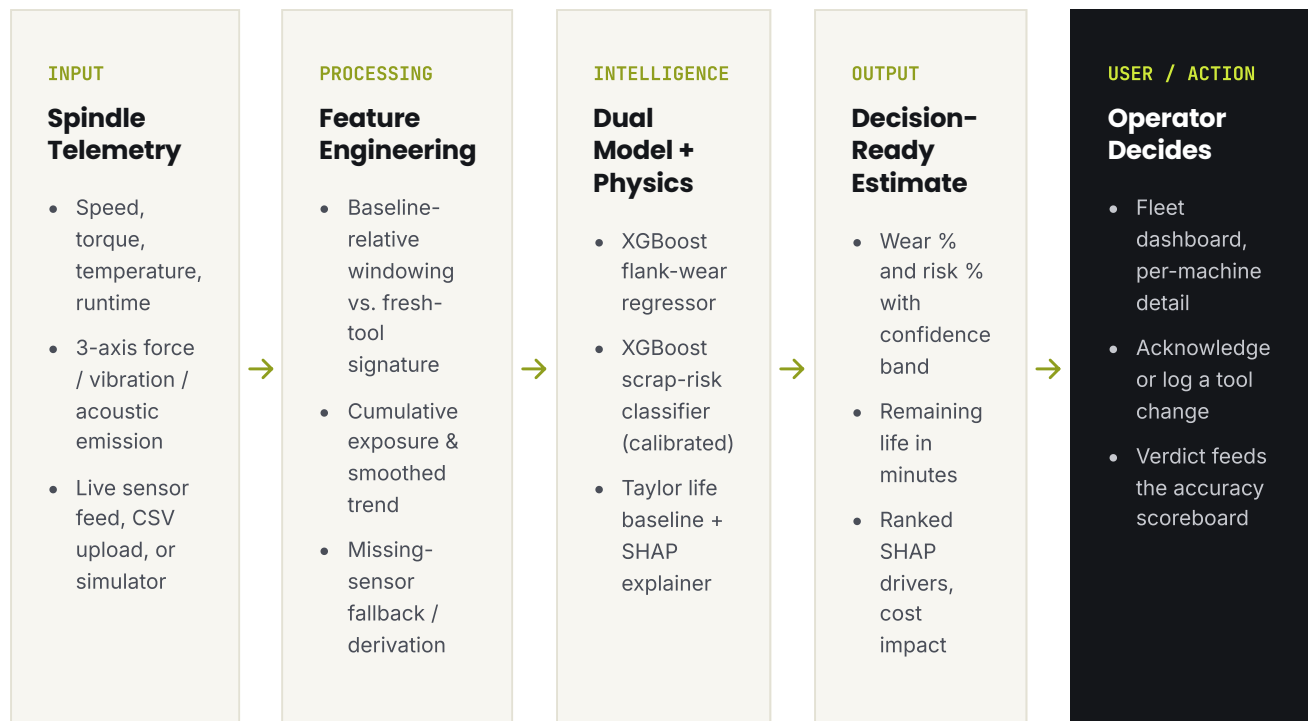
A Closed Human Feedback Loop

Every operator verdict on a tool change is logged and scored against what the model predicted, surfaced with a Wilson-score confidence interval — and honestly caveated as a field sanity check, not a validation metric, because it's a biased sample.

"The defensible claim is not 'better than any schedule' — it's 'removes the need to guess the schedule.'"

— from the project's own validation notes, on condition-based vs. fixed-interval tool changes

From spindle telemetry to a tool-change decision.



Never actuates the machine

Every endpoint in the system is read-and-recommend. Metrik does not stop, throttle, or reconfigure a machine — a person always makes the final call, and that call is what closes the feedback loop.

Persistent, multi-tenant fleet state

Every account's machines, tool-change history, and running scrap-saved total are mirrored to SQLite, so a server restart doesn't erase a plant's history. Two accounts never see each other's fleet.

One feature pipeline, two callers

The exact same feature-extraction code runs at training time and at inference time, so a live machine can never drift out of sync with how the model was actually trained — a common, quiet failure mode this design rules out by construction.

Graceful under an incomplete sensor package

A real shop's machines don't all report the same fields. Missing readings fall back to a training-median or are derived from related sensors — the system is designed to never crash on an incomplete payload.

What's actually running underneath.

BACKEND

Python · FastAPI

MACHINE LEARNING

XGBoost (regression + classification)

PERSISTENCE

SQLite — mirrors in-memory fleet state

CONVERSATIONAL ASSISTANT

Groq-hosted Llama 3.3 70B, rule-based fallback

FRONTEND

React 18 · Vite · TailwindCSS

EXPLAINABILITY

SHAP TreeExplainer, per prediction

AUTH

Bearer-token sessions, multi-tenant accounts

PHYSICS MODEL

Taylor tool-life equation, $V \cdot T^n = C$

Two Models, One Fleet

	SCRAP-RISK MODEL	FLANK-WEAR MODEL
Question	Probability this cut ends in a wear-driven scrap event	Flank wear right now, in millimetres
Training data	AI4I 2020 (synthetic benchmark)	PHM 2010 milling challenge (real, physically measured)
Validation	5×10 repeated stratified cross-validation	Leave-one-tool-out (trained on 2 cutters, tested on a 3rd)
Headline result	ROC-AUC 0.983 · PR-AUC 0.702	MAE 16.0 μm · R ² 0.724
Output	Calibrated probability (Platt scaling)	Millimetres, against the ISO 8688 0.30mm limit

Feature engineering

Seven condition-monitoring channels (3-axis force, 3-axis vibration, acoustic emission) are windowed every 2 seconds and expressed *relative to a fresh-tool baseline* captured in the first ~30 seconds after each tool change — because absolute force levels depend on the fixture and machine, not just the wear.

Robustness by design

CSV uploads are capped in size and row count against resource exhaustion; every sensor field has a documented fallback; the frontend badges any value the backend had to estimate rather than presenting it as measured.

API surface

Auth (register/login/session) · Fleet (machine list, detail, add/remove, state) · Predict (live & What-If simulator, CSV batch) · Human-in-the-loop (tool-change log, acknowledge, feedback scoreboard) · Chat · Export (JSON/PDF) — every fleet route scoped to the logged-in account.

Calibration, not just accuracy

The risk classifier's raw score is corrected with Platt scaling so a "70%" actually means observed at ~70% frequency — cutting the calibration error (Brier score) by 32%, without changing which machines alarm.

Verified against held-out data, not asserted.

16.0_{µm}

Flank-wear MAE, leave-one-tool-out — tested on a cutter the model never trained on.

0.724

R^2 for the wear regressor under the same held-out-tool validation.

0.983

ROC-AUC for the scrap-risk classifier (0.702 PR-AUC vs. a 0.012 random baseline).

20_{sec}

Remaining-life median error in the final quarter of tool life — when the answer matters most.

10.7×

optimism gap

The single most common error in tool-wear research is validating on a random row split — consecutive readings from the same cutter are nearly identical, so a model can memorize one tool's own wear curve. Measured on Metrik's exact data, that invalid split reports **1.5 µm MAE (R^2 0.995)** — a result **10.7× more optimistic** than the honest, leave-one-tool-out number of 16.0 µm. Both numbers are computed and reported every training run, on purpose.

Tool life recovered vs. a fixed schedule

Across the three validated cutters, a practical fixed schedule (built with an 80% safety margin) uses 74.1% of available tool life before changing it; Metrik's condition-based estimate uses 88.5% — **+19.3%** more usable life recovered, and **−4.6%** against an oracle schedule that would need foreknowledge no real shop has. Zero tools were changed late under either policy.

Honest limitation: cross-material transfer

Trained on steel and tested zero-shot on titanium cutters it never saw, wear-magnitude accuracy degrades from 16.0 µm to 38 µm MAE. Per-machine baselining is what makes deployment on an unseen machine viable at all — but the gap is measured and reported, not smoothed over.

All figures above are offline validation results on published research datasets (PHM 2010 Data Challenge; AI4I 2020 / Kaggle). The demo fleet replays these real, physically measured cutting signals rather than synthesizing them. The system has not yet been field-validated at a live production shop.

From a model to a shop-floor decision.

Live Fleet Dashboard

Per-machine wear/risk gauges, live sensor readout, and a fleet-wide KPI view polling continuously.

What-If Simulator

Move spindle speed, torque, depth-of-cut or runtime and watch the risk gauge and SHAP drivers respond live.

SHAP Driver View

Every prediction ships with a ranked breakdown of which sensor reading is actually driving the number.

CSV Batch Prediction

Upload a spreadsheet of readings and get a wear/risk verdict per row — no pre-loaded numbers, real model response.

Tool-Change Workflow

An operator logs whether a tool was genuinely worn when changed — a labelled event that closes the ML loop.

Model Feedback Scoreboard

Aggregates operator verdicts into a live accuracy read, with a statistically honest confidence interval.

Conversational Assistant

Ask about a machine's status in plain language; answered by a hosted LLM grounded in that machine's live data.

Multi-Tenant Accounts

Every plant's machines, history and scrap-saved total are scoped to its own account and persist across restarts.

WHO BENEFITS

Operators get a live, explained wear reading at the machine instead of a guess from a calendar. **Plant managers** get the fleet-wide view — which machines are at risk now, and evidence to plan maintenance windows around production.

WHAT IT REDUCES

Waste in *both* directions: parts scrapped by an undetected worn tool, and tools discarded early with usable carbide life still in them — the two costs a fixed schedule can never simultaneously avoid.

WHY IT SCALES DOWN-MARKET

Built on standard telemetry — speed, torque, temperature, runtime — rather than proprietary sensor hardware, so small and mid-sized job shops can adopt it without a vendor monitoring-suite budget.

Where this stands, and where it goes.

Why It Stands Out

- Grounds machine learning against a trusted physics equation instead of shipping a pure black box.
- Reports its own validation-leakage gap (10.7×) rather than quoting the flattering number.
- Explains every prediction with SHAP; never hands over a score with no reason attached.
- Closes the loop with real operator verdicts, honestly caveated for sampling bias.

Current Implementation

WORKING PROTOTYPE

- Full-stack app: FastAPI backend, React dashboard, SQLite persistence, multi-tenant auth.
- Both models trained, calibrated, and leave-one-tool-out / cross-validated with reported metrics.
- CSV batch prediction, What-If simulator, tool-change workflow, and feedback scoreboard all working end-to-end.

Current Prototype Scope

HONEST LIMITS

- Demo telemetry replays real recorded PHM 2010 signals; not yet wired to a live shop-floor sensor feed.
- The risk classifier trains on a synthetic benchmark, not field failure data — stated plainly, not hidden.
- Not yet pilot-tested with an external shop; no field deployment exists today.

Future Potential

NEXT STEPS

- Live sensor / PLC integration with a partner machine shop for a real pilot.
- Extend wear-model generalization across work materials beyond steel.
- Mobile / edge-side inference for shop floors without a fixed workstation.

Metrik doesn't ask a shop to trust a black box — it grounds a machine-learning wear estimate in the physics the industry already trusts, explains every number it shows, and reports its own validation honestly enough to survive scrutiny. That combination, not any single model, is the innovation.

